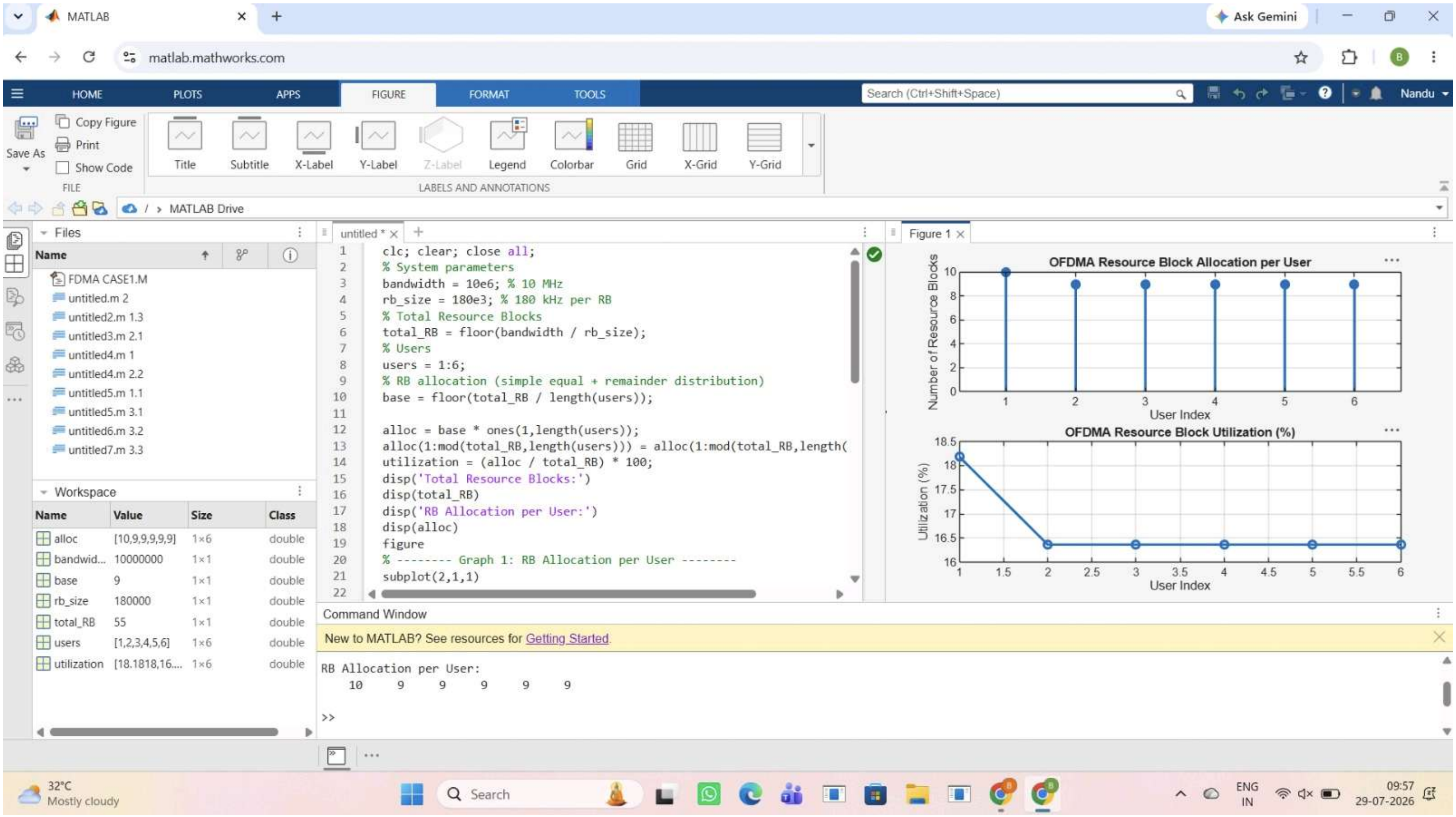
MATLAB
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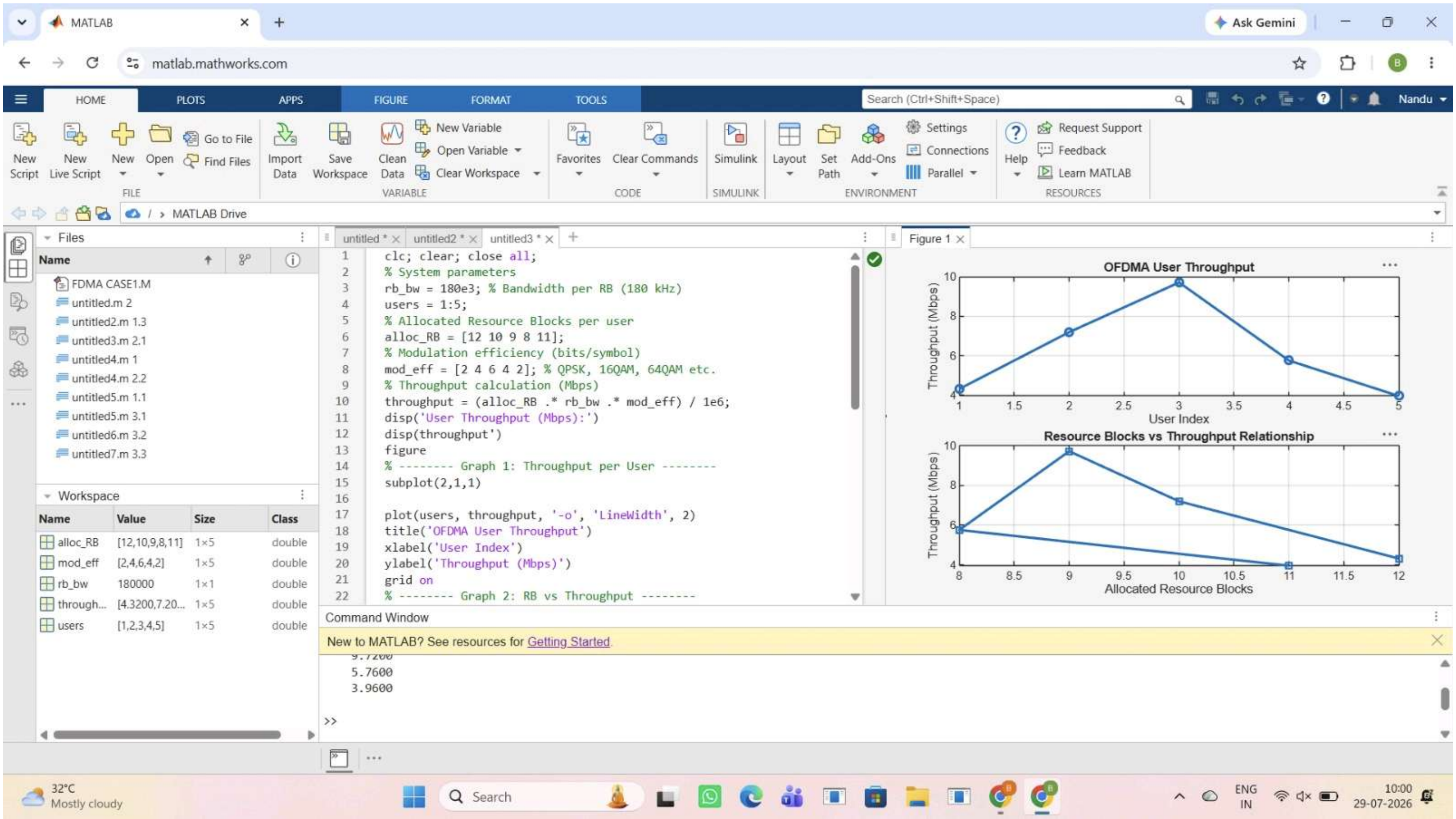
HOME PLOTS APPS FIGURE FORMAT TOOLS
MATLAB Drive
Files
FDMA CASE1.M
untitled.m 2
untitled2.m 1.3
untitled3.m 2.1
untitled4.m 1
untitled4.m 2.2
untitled5.m 1.1
untitled5.m 3.1
untitled6.m 3.2
untitled7.m 3.3
Workspace
Name Value Size Class
alloc [10,9,9,9,9,9] 1×6 double
bandwid... 10000000 1×1 double
base 9 1×1 double
rb_size 180000 1×1 double
total_RB 55 1×1 double
users [1,2,3,4,5,6] 1×6 double
utilization [18.1818,16.... 1×6 double
clc; clear; close all;
% System parameters
bandwidth = 10e6; % 10 MHz
rb_size = 180e3; % 180 kHz per RB
% Total Resource Blocks
total_RB = floor(bandwidth / rb_size);
% Users
users = 1:6;
% RB allocation (simple equal + remainder distribution)
base = floor(total_RB / length(users));
alloc = base * ones(1,length(users));
alloc(1:mod(total_RB,length(users))) = alloc(1:mod(total_RB,length(
utilization = (alloc / total_RB) * 100;
disp('Total Resource Blocks:')
disp(total_RB)
disp('RB Allocation per User:')
disp(alloc)
figure
% -------- Graph 1: RB Allocation per User --------
subplot(2,1,1)
OFDMA Resource Block Allocation per User
Number of Resource Blocks
User Index
OFDMA Resource Block Utilization (%)
Utilization (%)
User Index
Command Window
New to MATLAB? See resources for Getting Started.
RB Allocation per User:
10 9 9 9 9 9
32°C Mostly cloudy
09:57 29-07-2026



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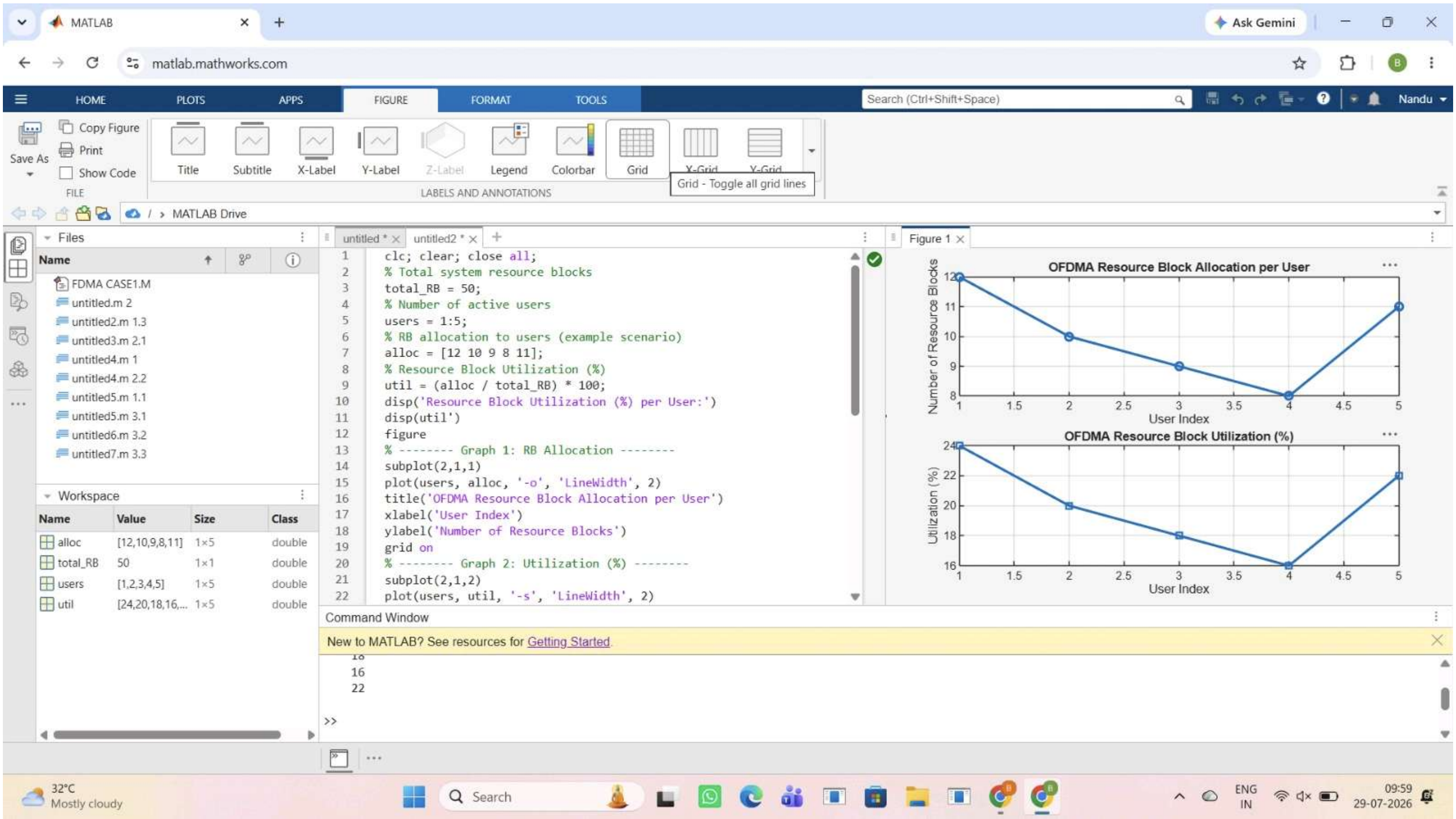
FDMA CASE1.M
Workspace
untitled * untitled2 * untitled3 *
clc; clear; close all;
% System parameters
rb_bw = 180e3; % Bandwidth per RB (180 kHz)
users = 1:5;
% Allocated Resource Blocks per user
alloc_RB = [12 10 9 8 11];
% Modulation efficiency (bits/symbol)
mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
% Throughput calculation (Mbps)
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
disp('User Throughput (Mbps):')
disp(throughput')
figure
% -------- Graph 1: Throughput per User --------
subplot(2,1,1)
plot(users, throughput, '-o', 'LineWidth', 2)
title('OFDMA User Throughput')
xlabel('User Index')
ylabel('Throughput (Mbps)')
grid on
% -------- Graph 2: RB vs Throughput --------
Figure 1
OFDMA User Throughput
Throughput (Mbps)
User Index
Resource Blocks vs Throughput Relationship
Throughput (Mbps)
Allocated Resource Blocks
Command Window
New to MATLAB? See resources for Getting Started.
5.7600
3.9600
>>



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Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
Grid - Toggle all grid lines
LABELS AND ANNOTATIONS
MATLAB Drive
Files
Name
FDMA CASE1.M
untitled.m 2
untitled2.m 1.3
untitled3.m 2.1
untitled4.m 1
untitled4.m 2.2
untitled5.m 1.1
untitled5.m 3.1
untitled6.m 3.2
untitled7.m 3.3
Workspace
Name Value Size Class
alloc [12,10,9,8,11] 1×5 double
total_RB 50 1×1 double
users [1,2,3,4,5] 1×5 double
util [24,20,18,16,... 1×5 double
untitled *
untitled2 *
clc; clear; close all;
% Total system resource blocks
total_RB = 50;
% Number of active users
users = 1:5;
% RB allocation to users (example scenario)
alloc = [12 10 9 8 11];
% Resource Block Utilization (%)
util = (alloc / total_RB) * 100;
disp('Resource Block Utilization (%) per User:')
disp(util')
figure
% -------- Graph 1: RB Allocation --------
subplot(2,1,1)
plot(users, alloc, '-o', 'LineWidth', 2)
title('OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on
% -------- Graph 2: Utilization (%) --------
subplot(2,1,2)
plot(users, util, '-s', 'LineWidth', 2)
Figure 1
OFDMA Resource Block Allocation per User
Number of Resource Blocks
User Index
OFDMA Resource Block Utilization (%)
Utilization (%)
User Index
Command Window
New to MATLAB? See resources for Getting Started.
16
22
>>
32°C
Mostly cloudy
Search
ENG
IN
09:59
29-07-2026